

or any wearable device to continuously monitor and generate feedback. Surgical equipment widely use these for better results.

Automobiles. MEMS-based accelerometers and gyroscopes are placed in different parts of a car for a variety of applications. These sensors are ideal for their unique combination of stability, reliability, high sensitivity, low cost, compact size, wide range and high quality. Most of the European car manufacturers have adopted MEMS accelerometers in their designs. These sensors give digital outputs that can be fed to the central microprocessor for it to perform the complex reactive functions.

"The airbag control unit analyses data regarding acceleration and angular rates to decide whether the restraint systems are to be activated. They provide the physical basis for an airbag system that works. In terms of size, weight and cost, MEMS sensors are state-of-the-art. Currently we see no alternative technology for an airbag application," says Matthias Brock, head of media relations, R&D and environmental communications, Daimler AG, Germany. "Depending upon which system is used, they are mounted at vibration-free structures in the interior and the engine-bay," he adds.

A recent collaboration between Analog Devices, the largest MEMS supplier for automobiles and Infineon, a supplier for application-specific components for automobiles, promises customers a complete, interoperable and validated airbag system platform. This will accelerate development of advanced, next-generation automotive safety systems.

The tilt sensor in electronic parking brake systems automatically activates the parking brakes to prevent the slipping of the car. A computerised system based on motion sensors and actuators, called Electronic Stability Control (ESC), is used to enhance the safety in vehicles. In the BMW X5, the MEMS sensors collect data while the car is being driven and signals the suspension

References

1. Girish Krishnan, Chaitanya U. Kshirsagar, G.K. Ananthasuresh and Navakanta Bhat's 'Micromachined high resolution accelerometers', *Journal of Indian Institute of Science*, Vol 87:3, Jul-Sep 2007
2. Yole Development report on 'MEMS Accelerometer, gyroscope and IMU Market 2008-2013', June 2009
3. InvenSense Inc. documentation on 'Selection and Integration of MEMS based motion processing in consumer apps' by Steve Nasiri, David Sachs and Michael Maia

and handling to adjust automatically. MEMS-based pressure sensors are also used in automobile combustion systems and in power trains.

Consumer Electronics. Vivid forms of motion can be useful in a variety of consumer electronics goods. Industry leaders in this domain have realised the potential in MEMS-based accelerometers and are now tapping the market with innovative applications. Music players from Sony use the 'shake-up' feature to shuffle the songs. Gestures are also used to scroll and go to the next playlist. Laptops, cameras, MP3 players, digital compasses, etc, have some new features that use motion sensing innovatively.

Challenges for MEMS designers

MEMS-based chip designers are working towards getting robust and high-performance single-package solutions. "The major challenge in MEMS is the bonding between various ICs inside a chip. As MEMS is a combination of mechanical and electrical structures, the wire bonding between the chips has to be done in a robust way within a minute space," says Sapna Mongia, technical marketing manager, STMicroelectronics. The principle of sensing is different in linear and rotational motion, which is another challenge when combining all into a single chip.

Digital signal processing and micro-fabrication are the key drivers in developing future MEMS sensors.

MEMS is still a nascent technology and there is great potential for growth. "MEMS face many challenges with respect to design and fabrication, but the biggest challenge I see is the lack of expertise in the field of MEMS," reports Raman Iyer, technical sales, FTD Infocom.

MEMS designers and product designers make use of various software and tools for an efficient system design. Software like Coventor MEMS+, which is a design automation software, can be used to simulate the behaviour of the design and can generate an automatic parametrised physical layout. Cadence is a design automation tool for the custom IC industry. MEMS simulation models and access to the MEMS design library helps in better design concepts.

From supplying only the MEMS chip, many MEMS players are moving to supplying sub-assemblies that can be integrated with the product. MEMS manufacturers work in conjunction with other companies in order to develop application-specific MEMS motion sensors. They also provide evaluation boards that come with software for the application designer.

MEMS business continues to be driven by innovation. Inertial sensors represent serious business, according to Yole Development's report. Batch processing techniques for micro-fabrication significantly lower the cost of production and produce large volumes of uniform and highly reliable devices. Nearly 750 million units of MEMS accelerometers and gyroscopes were manufactured in 2008, which is a \$1.8 billion market already. A major part of the business still comes from automobile applications, but consumer electronics (especially mobile phones) is seen as potentially the biggest growth driver for MEMS devices. MEMS-based products will continue to be in demand for their high-end user interface and motion sensing that can be applied innovatively to novel applications. ●

The author is senior journalist, technology, at EYF